

Slam- Video Navigation - Exercise 3

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Git Repository: github.com/yehonatan1305/SLAM-VN

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Question 3

- For every $w = \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix}$ in left 0 coordinates, we define the transformation to left1 coordinates as:

$$T(w) = R \cdot w + t$$

- The transformation from camera A to camera C is:

$$T_{A \rightarrow C}(x) = T_{B \rightarrow C}(T_{A \rightarrow B}(x)) = R_2 R_1 x + R_2 t_1 + t_2$$

The extrinsic matrix of C relative to A is:

$$[R_2 R_1 \mid R_2 t_1 + t_2]$$

- As we discussed in class, the camera center c in the world coordinate system is the point that is mapped to the origin $(0,0,0)$ in the camera coordinate system by the extrinsic matrix. That is, if the extrinsic matrix is $[R \mid t]$, then the camera center c satisfies:

$$[R \mid t] \begin{pmatrix} c \\ 1 \end{pmatrix} = 0$$

This constrain leads us to the equation $Rc + t = 0$, so $c = -R^{-1}t = -R^T t$ (since R is a rotation matrix and $R^{-1} = R^T$).

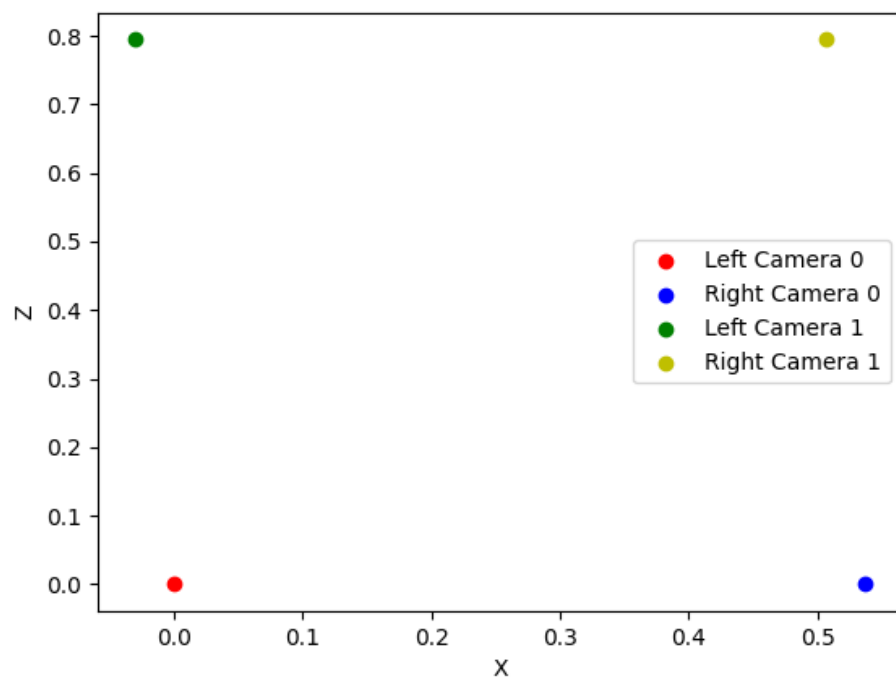


Figure 1: Top-down view of the estimated positions of the four cameras in the global coordinate system. (ex3.py line 46)

Question 4

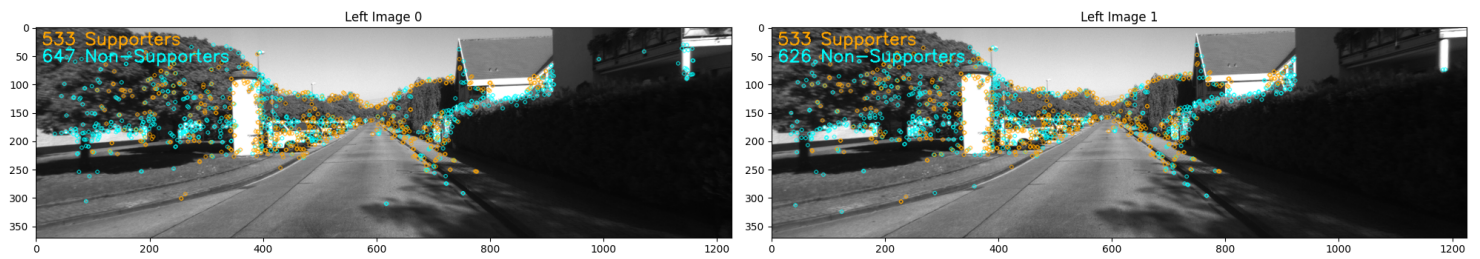


Figure 2: Visualization of keypoint matches between left0 and left1. Supporters are highlighted in blue. (ex3.py line 61)

Question 5

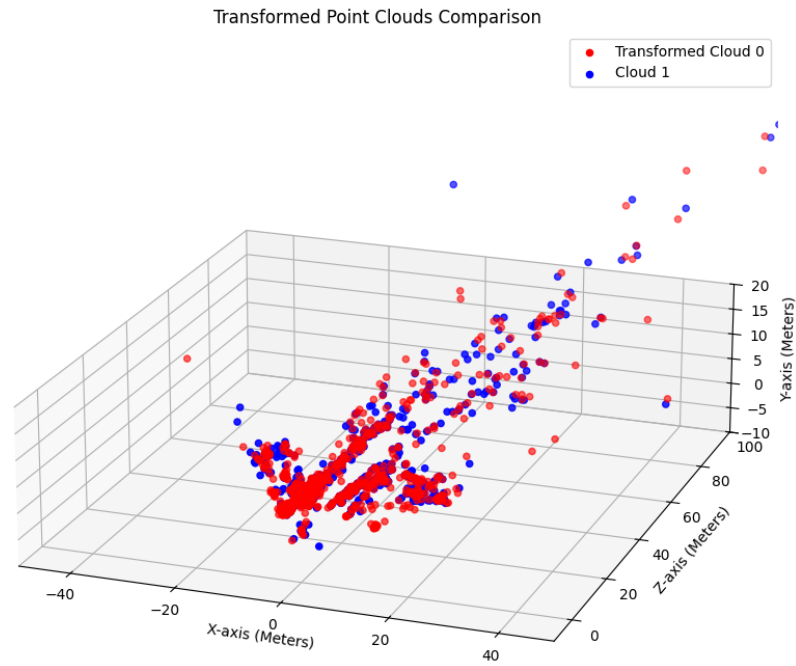


Figure 3: Top-down view of the two 3D point clouds after alignment. (ex3.py line 93)

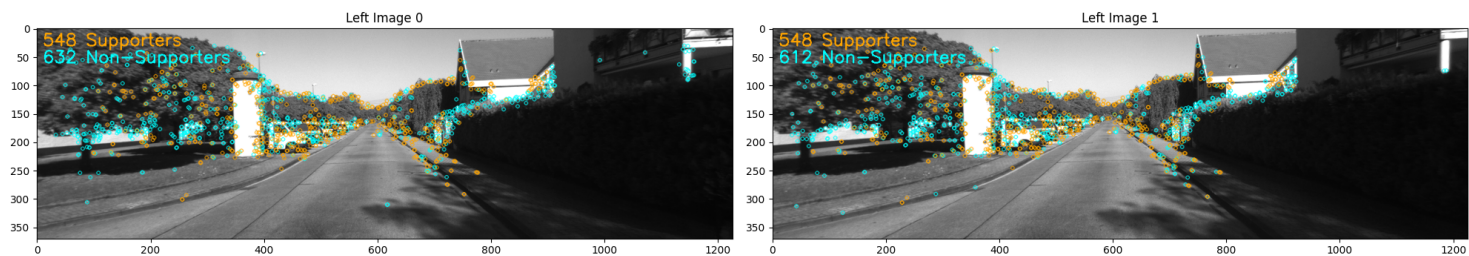


Figure 4: Inlier and outlier matches on the images, as determined by Ransac-PnP. (ex3.py line 108)

Question 6

The total time of tracking is 556.3538 seconds.

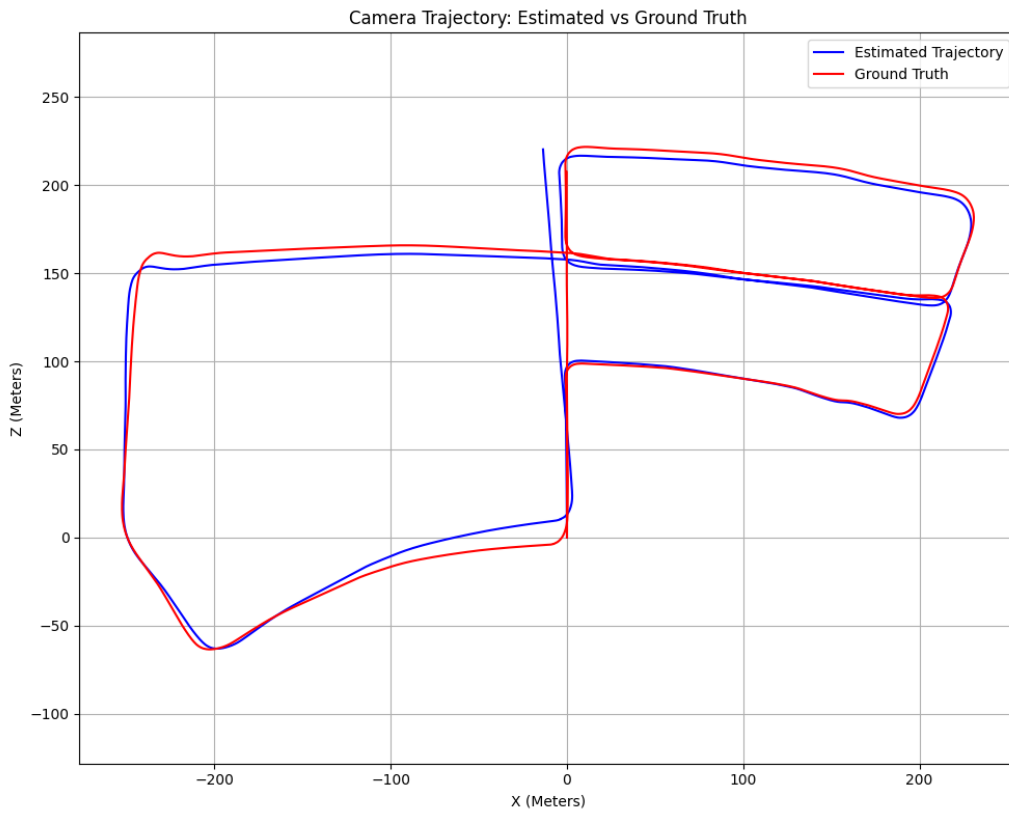


Figure 5: Estimated camera trajectory vs. ground truth in the global coordinate system. (q6.py line 110)